THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of x in $7x + 2y = -6$, $3x + 5y = 14$ is: 1

(a) 4 **(b)** -8 **(c)** -2 **(d)** 2

2. The value of y in $11x + 4y = 53$, $5x + 3y = 30$ is: 1

(a) 8 **(b)** -5 **(c)** 5 **(d)** 3

3. The pair $2x + ky = 6$, $4x + 4y = 12$ has infinitely many solutions when $k =$ 1

(a) 6 **(b)** 3 **(c)** 2 **(d)** 4

4. For the pair $kx + 2y = 5$ and $6x + y = 3$ to have a unique solution, k must NOT equal: 1

(a) 12 **(b)** 3 **(c)** 10 **(d)** 8

5. If $3x + 2y = 25$ and $2x + 3y = 20$, then $x + y =$ 1

(a) 5 **(b)** 45 **(c)** 9 **(d)** $\frac{45}{2}$

6. If $x + y = 9$ and $x - y = 3$, then $x^2 - y^2 =$ 1

(a) 27 **(b)** 12 **(c)** 72 **(d)** 90

7. The value of x in $5x + 3y = 29$, $4x + 7y = 37$ is: 1

(a) 7 **(b)** 4 **(c)** -4 **(d)** 3

8. The value of y in $13x + 5y = 67$, $6x + 7y = 45$ is: 1

(a) 7 **(b)** -3 **(c)** 4 **(d)** 3

9. The pair $3x + ky = 9$, $6x + 6y = 18$ has infinitely many solutions when $k =$ 1

(a) 9 **(b)** 6 **(c)** 3 **(d)** 4

10. For the pair $kx + 3y = 4$ and $8x + 2y = 5$ to have a unique solution, k must NOT equal: 1

(a) $\frac{16}{3}$ **(b)** 11 **(c)** 12 **(d)** $\frac{32}{5}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve by the cross-multiplication method: $7x + 2y = -6$ and $3x + 5y = 14$. 2

12. For what value of k does the pair $kx + 2y = 5$, $6x + y = 3$ have a unique solution? 2

13. Find k so that the pair $2x + ky = 6$ and $4x + 4y = 12$ has infinitely many solutions. 2

14. Solve by the cross-multiplication method: $5x + 3y = 29$ and $4x + 7y = 37$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by elimination: $7x + 2y = -6$ and $3x + 5y = 14$. **3**

16. The sum of the digits of a two-digit number is 11. If the digits are interchanged, the number becomes 45 more. Find the number. **3**

17. Find the value of k for which the pair $kx + y = k - 1$ and $4x + ky = k$ has infinitely many solutions. **3**

OR

Solve by elimination: $5x + 3y = 29$ and $4x + 7y = 37$. **3**

18. The sum of the digits of a two-digit number is 13. If the digits are interchanged, the number becomes 45 more. Find the number. **3**

OR

Find the value of k for which the pair $kx + 2y = k - 2$ and $8x + ky = k$ has infinitely many solutions. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A father is 10 times as old as his son. After 6 years the father will be 4 times as old as the son will be. Find their present ages. **5**

OR

A fraction becomes $\frac{1}{2}$ when 1 is added to both the numerator and the denominator, and becomes $\frac{5}{9}$ when 2 is added to both. Find the fraction. **5**

20. A father is 6 times as old as his son. After 10 years the father will be 5 times as old as the son will be. Find their present ages. **5**

OR

A fraction becomes $\frac{2}{3}$ when 3 is added to both the numerator and the denominator, and becomes $\frac{11}{15}$ when 6 is added to both. Find the fraction. **5**

END OF QUESTION PAPER — Total: 40 Marks

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